

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 25: Probability

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Probability of an event

1. Which of these cannot be the probability of an event? (a) $\frac{5}{4}$ (b) 0.25 (c) $\frac{1}{33}$ (d) 67%

ICSE 2025, 1 mark

SOLUTION

A probability always lies from 0 to 1. $\frac{5}{4} = 1.25$ is greater than 1.

Answer: (a)

2 Balls, numbered cards and letters

2. A box holds cards numbered 1 to 30, and one card is drawn at random. A wall clock is won for a perfect square, a water bottle for an even number that is also a multiple of 3, and a purse for a prime number. Find the probability of winning (a) the wall clock (b) the water bottle (c) the purse.

ICSE 2025, 3 marks

SOLUTION

There are 30 equally likely outcomes.

(a) Perfect squares: 1, 4, 9, 16, 25 (5 cards). Probability = $\frac{5}{30} = \frac{1}{6}$.

(b) Even multiples of 3 are the multiples of 6: 6, 12, 18, 24, 30 (5 cards). Probability = $\frac{5}{30} = \frac{1}{6}$.

(c) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 (10 cards). Probability = $\frac{10}{30} = \frac{1}{3}$.

Answer: (a) $\frac{1}{6}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$

3. The letters A, D, M, N, O, S, U, Y are written on separate cards. One card is drawn at random. Find the probability that the letter (a) is in the word MONDAY (b) is not in MONDAY (c) is in both SUNDAY and MONDAY. **ICSE 2024, 3 marks**

SOLUTION

There are 8 cards.

(a) Letters of MONDAY: M, O, N, D, A, Y, which is 6 cards. Probability = $\frac{6}{8} = \frac{3}{4}$.

(b) $1 - \frac{3}{4} = \frac{1}{4}$ (the letters S and U).

(c) Common to both words: N, D, A, Y, which is 4 cards. Probability = $\frac{4}{8} = \frac{1}{2}$.

Answer: (a) $\frac{3}{4}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$

3 Coins and dice

4. Assertion (A): When a die is thrown once, the probability of getting an even number is $\frac{2}{3}$. Reason (R): The even numbers on a die are {2, 4, 6}. (a) A true, R false (b) A false, R true (c) both true (d) both false **ICSE 2024, 1 mark**

SOLUTION

There are 3 even numbers out of 6, so the probability is $\frac{3}{6} = \frac{1}{2}$. A is false.

R is true.

Answer: (b)

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